

Gilbert Strang - A First Course in Calculus

Chapters 1 - 7

1 Chapter 1: Introduction to Vectors

1.1 Vectors and Linear Combinations

Problems 1-9 are about addition of vectors and linear combinations.

1. Describe geometrically (line, plane, or all of R^3) all linear combinations of

(a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

2. Draw $v = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $v + w$ and $v - w$ in a single xy plane.

3. If $v + w = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $v - w = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, compute and draw the vectors v and w .

4. From $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, find the components of $3v + w$ and $cv + dw$.

5. Compute $u + v + w$ and $2u + 2v + w$. How do you know u, v, w lie in a plane?

These lie in a plane because
 $w = cu + dv$. Find c and d

$$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v = \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}, \quad w = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}.$$

6. Every combination of $v = (1, -2, 1)$ and $w = (0, 1, -1)$ has components that add to _____.
Find c and d so that $cv + dw = (3, 3, -6)$. Why is $(3, 3, 6)$ impossible?

7. In the xy plane mark all nine of these linear combinations:

$$c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{with} \quad c = 0, 1, 2 \quad \text{and} \quad d = 0, 1, 2.$$

8. The parallelogram in Figure 1.1 has diagonal $v + w$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.

9. If three corners of a parallelogram are $(1, 1)$, $(4, 2)$, and $(1, 3)$, what are all three of the possible fourth corners? Draw two of them.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. Which point of the cube is $i + j$? Which point is the vector sum of $i = (1, 0, 0)$ and $j = (0, 1, 0)$ and $k = (0, 0, 1)$? Describe all points (x, y, z) in the cube.
11. Four corners of this unit cube are $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are _____. The cube has how many edges?
12. Review Question. In xyz space, where is the plane of all linear combinations of $i = (1, 0, 0)$ and $i + j = (1, 1, 0)$?

13. (a) What is the sum V of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?
- (b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?
- (c) What are the x, y components of that 2:00 vector $v = (\cos \theta, \sin \theta)$?
14. Suppose the twelve vectors start from 6:00 at the bottom instead of $(0, 0)$ at the center. The vector to 12:00 is doubled to $(0, 2)$. The new twelve vectors add to _____.

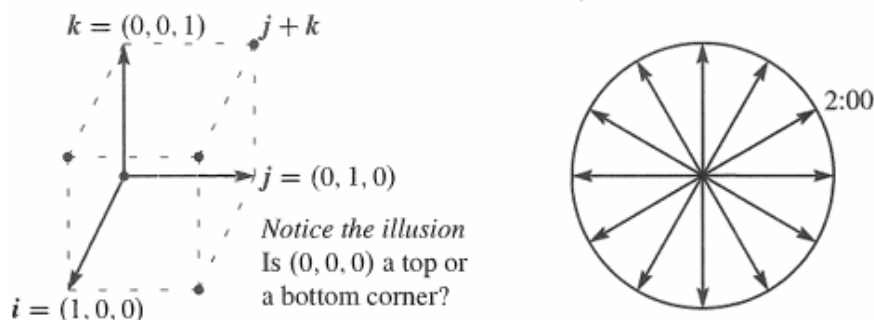


Figure 1.4: Unit cube from i, j, k and twelve clock vectors.

Problems 15–19 go further with linear combinations of v and w (Figure 1.5a).

15. Figure 1.5a shows $\frac{1}{2}v + \frac{1}{2}w$. Mark the points $\frac{3}{4}v + \frac{1}{4}w$ and $\frac{1}{4}v + \frac{3}{4}w$ and $v + w$.
16. Mark the point $-v + 2w$ and any other combination $cv + dw$ with $c + d = 1$. Draw the line of all combinations that have $c + d = 1$.
17. Locate $\frac{1}{3}v + \frac{1}{3}w$ and $\frac{2}{3}v + \frac{2}{3}w$. The combinations $cv + cw$ fill out what line?
18. Restricted by $0 \leq c \leq 1$ and $0 \leq d \leq 1$, shade in all combinations $cv + dw$.
19. Restricted only by $c \geq 0$ and $d \geq 0$ draw the “cone” of all combinations $cv + dw$.

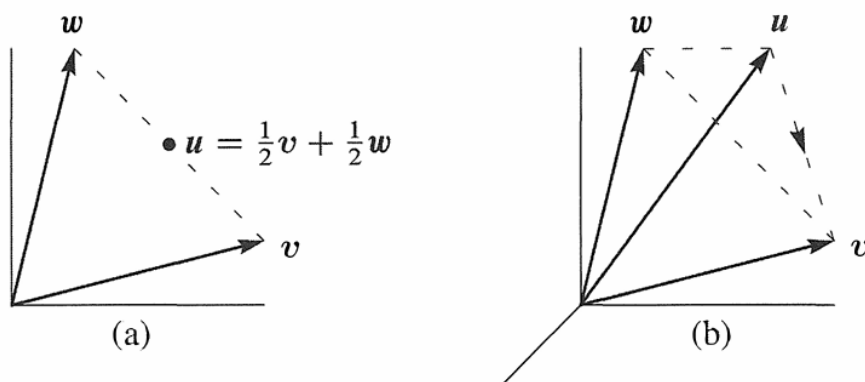


Figure 1.5: Problems 15-19 in a plane

Problems 20-25 in 3-dimensional space

Problems 20-25 deal with u, v, w in three-dimensional space (see Figure 1.5b).

20. Locate $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}v$ and $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}w$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e , will the combinations $cu + dv + ew$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.
21. The three sides of the dashed triangle are $v - u$ and $w - v$ and $u - w$. Their sum is _____. Draw the head-to-tail addition around a plane triangle of $(3, 1)$ plus $(-1, 1)$ plus $(-2, -2)$.

22. Shade in the pyramid of combinations $cu + dv + ew$ with $c \geq 0, d \geq 0, e \geq 0$ and $c + d + e \leq 1$. Mark the vector $\frac{1}{2}(u + v + w)$ as inside or outside this pyramid.
23. If you look at all combinations of those u, v , and w , is there any vector that can't be produced from $cu + dv + ew$? Different answer if u, v, w are all in _____.
24. Which vectors are combinations of u and v , and also combinations of v and w ?
25. Draw vectors u, v, w so that their combinations $cu + dv + ew$ fill only a line. Find vectors u, v, w so that their combinations $cu + dv + ew$ fill only a plane.
26. What combination $c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ produces $\begin{bmatrix} 14 \\ 8 \end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

Challenge Problems

27. How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0, 0, 1, 0)$. A typical edge goes to $(0, 1, 0, 0)$.
28. Find vectors v and w so that $v + w = (4, 5, 6)$ and $v - w = (2, 5, 8)$. This is a question with _____ unknown numbers, and an equal number of equations to find those numbers.
29. Find two different combinations of the three vectors $u = (1, 3)$ and $v = (2, 7)$ and $w = (1, 5)$ that produce $b = (0, 1)$. Slightly delicate question: If I take any three vectors u, v, w in the plane, will there always be two different combinations that produce $b = (0, 1)$?
30. The linear combinations of $v = (a, b)$ and $w = (c, d)$ fill the plane unless _____. Find four vectors u, v, w, z with four components each so that their combinations $cu + dv + ew + fz$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.
31. Write down three equations for c, d, e so that $cu + dv + ew = b$. Can you somehow find c, d, e for b ?

$$u = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

1.2 Lengths and Dot Products

1. Calculate the dot products $u \cdot v$ and $u \cdot w$ and $u \cdot (v + w)$ and $w \cdot v$:

$$u = \begin{bmatrix} -6 \\ .8 \end{bmatrix}, \quad v = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

2. Compute the lengths $\|u\|$ and $\|v\|$ and $\|w\|$ of those vectors. Check the Schwarz inequalities $|u \cdot v| \leq \|u\| \|v\|$ and $v \cdot w \leq \|v\| \|w\|$.
3. Find unit vectors in the directions of v and w in Problem 1, and the cosine of the angle θ . Choose vectors a, b, c that make $0^\circ, 90^\circ$, and 180° angles with w .
4. For any *unit* vectors v and w , find the dot products (actual numbers) of
- (a) v and $-v$ (b) $v + w$ and $v - w$ (c) $v - 2w$ and $v + 2w$
5. Find unit vectors u_1 and u_2 in the directions of $v = (1, 3)$ and $w = (2, 1, 2)$. Find unit vectors U_1 and U_2 that are perpendicular to u_1 and u_2 .
6. (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$.
 (b) All vectors perpendicular to $V = (1, 1, 1)$ lie on a _____ in 3 dimensions.
 (c) The vectors perpendicular to both $(1, 1, 1)$ and $(1, 2, 3)$ lie on a _____.
7. Find the angle θ (from its cosine) between these pairs of vectors:

$$(a) \ v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \text{ and } w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(b) \ v = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \text{ and } w = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$$(c) \ v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \text{ and } w = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

$$(d) \ v = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ and } w = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

8. True or false (give a reason if true or find a counterexample if false):

(a) If $u = (1, 1, 1)$ is perpendicular to v and w , then v is parallel to w .

(b) If u is perpendicular to v and w , then u is perpendicular to $v + 2w$.

(c) If u and v are perpendicular unit vectors then $\|u - v\| = \sqrt{2}$. Yes!

9. The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1 . Suppose the product v_2w_2/v_1w_1 of those slopes is -1 . Show that $v \cdot w = 0$ and the vectors are perpendicular. (The line $y = 4x$ is perpendicular to $y = -\frac{1}{4}x$.)

10. Draw arrows from $(0, 0)$ to the points $v = (1, 2)$ and $w = (-2, 1)$. Multiply their slopes. That answer is a signal that $v \cdot w = 0$ and the arrows are _____.

11. If $v \cdot w$ is negative, what does this say about the angle between v and w ? Draw a 3-dimensional vector v (an arrow), and show where to find all w 's with $v \cdot w < 0$.

12. With $v = (1, 1)$ and $w = (1, 5)$ choose a number c so that $w - cv$ is perpendicular to v . Then find the formula for c starting from any nonzero v and w .

13. Find nonzero vectors v and w that are perpendicular to $(1, 0, 1)$ and to each other.

14. Find nonzero vectors u, v, w that are perpendicular to $(1, 1, 1)$ and to each other.

15. The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) = \frac{1}{2}(2 + 8) = 5$. This would come in Example 6 from the Schwarz inequality for $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$. Find $\cos \theta$ for this v and w .

16. How long is the vector $v = (1, 1, \dots, 1)$ in 9 dimensions? Find a unit vector u in the same direction as v and a unit vector w that is perpendicular to v .

17. What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors i, j, k along the axes? Check the formula $\cos^2 \alpha + \cos^2 \beta + \cos^2 \theta = 1$.

Problems 18-28 lead to the main facts about lengths and angles in triangles.

18. The parallelogram with sides $v = (4, 2)$ and $w = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for *right triangles only*:

$$(\text{length of } v)^2 + (\text{length of } w)^2 = (\text{length of } v + w)^2.$$

19. (Rules for dot products) These equations are simple but useful:

$$(a) \ v \cdot w = w \cdot v$$

$$(b) \ u \cdot (v + w) = u \cdot v + u \cdot w$$

$$(c) \ (cv) \cdot w = c(v \cdot w)$$

Use (2) with $u = v + w$ to prove $\|v + w\|^2 = v \cdot v + 2v \cdot w + w \cdot w$.

20. The "Law of Cosines" comes from $(v - w) \cdot (v - w) = v \cdot v - 2v \cdot w + w \cdot w$:

$$\text{Cosine Law} \quad \|v - w\|^2 = \|v\|^2 - 2\|v\|\|w\|\cos \theta + \|w\|^2.$$

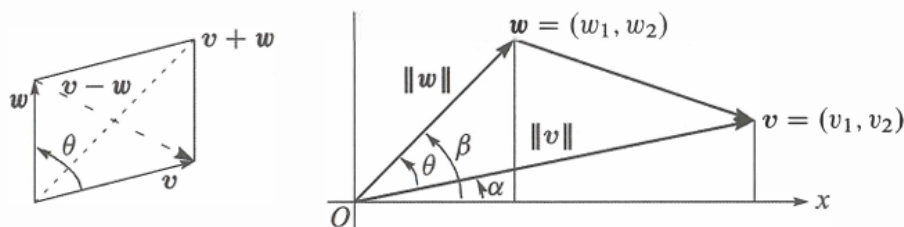
Draw a triangle with sides v and w and $v - w$. Which of the angles is θ ?

21. The triangle inequality says: (length of $v + w$) $\leq$ (length of v) + (length of w).

Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$.

Increase by $\|v\| \|w\|$ to show that $v \cdot w$ can not exceed $\|v\| \|w\|$:

Triangle inequality $\|v + w\|^2 \leq (\|v\| + \|w\|)^2$ or $\boxed{\|v + w\| \leq \|v\| + \|w\|}$.



22. The Schwarz inequality $|v \cdot w| \leq \|v\| \|w\|$ by algebra instead of trigonometry:

(a) Multiply out both sides of $(v_1 w_1 + v_2 w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.

(b) Show that the difference between those two sides equals $(v_1 w_2 - v_2 w_1)^2$. This cannot be negative since it is a square—so the inequality is true.

23. The figure shows that $\cos \alpha = v_1/\|v\|$ and $\sin \alpha = v_2/\|v\|$. Similarly $\cos \beta$ is _____ and $\sin \beta$ is _____. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos \beta \cos \alpha + \sin \beta \sin \alpha$ for $\cos(\beta - \alpha)$ to find $\cos \theta = v \cdot w / \|v\| \|w\|$.

24. One-line proof of the inequality $|u \cdot U| \leq 1$ for unit vectors (u_1, u_2) and (U_1, U_2) :

$$|u \cdot U| \leq |u_1| |U_1| + |u_2| |U_2| \leq \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos \theta$.

25. Why is $|\cos \theta|$ never greater than 1 in the first place?

26. (Recommended) Draw a parallelogram.

27. Parallelogram with two sides v and w . Show that the squared diagonal lengths $\|v + w\|^2 + \|v - w\|^2$ add to the sum of four squared side lengths $2\|v\|^2 + 2\|w\|^2$.

28. If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy -plane with $v \cdot w = x + 2y = 5$. Why do those w 's lie along a line? Which is the shortest w ?

29. (Recommended) If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest possible values of $\|v - w\|$? What are the smallest and largest possible values of $v \cdot w$?

Challenge Problems

30. Can three vectors in the xy -plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$?

I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible....)

31. Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w / \|v\| \|w\|$ is always $-\frac{1}{2}$.

32. How could you prove $\sqrt[3]{xyz} \leq \frac{1}{3}(x + y + z)$ (geometric mean $\leq$ arithmetic mean)?

33. Find 4 perpendicular unit vectors of the form $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2})$: Choose + or -.

34. Using $v = \text{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v/\|v\|$.

Using $V = \text{randn}(3, 30)$ create 30 more random unit vectors U_j .

What is the average size of the dot products $|u \cdot U_j|$?

In calculus, the average is $\int_0^\pi |\cos \theta| d\theta / \pi = 2/\pi$.

1.3 Matrices

1. Find the linear combination $3s_1 + 4s_2 + 5s_3 = b$. Then write b as a matrix-vector multiplication Sx , with 3, 4, 5 in x . Compute the three dot products (row of S) $\cdot x$:

$$s_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad s_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{go into the columns of } S.$$

2. Solve these equations $Sy = b$ with s_1, s_2, s_3 in the columns of S :

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}.$$

S is a sum matrix. The sum of the first 3 odd numbers is _____.

3. Solve these three equations for y_1, y_2, y_3 in terms of c_1, c_2, c_3 :

$$Sy = c \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Write the solution y as a matrix $A = S^{-1}$ times the vector c . Are the columns of S independent or dependent?

4. Find a combination $x_1w_1 + x_2w_2 + x_3w_3$ that gives the zero vector with $x_1 = 1$:

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

Those vectors are (independent) (dependent). The three vectors lie in a _____. The matrix W with those three columns is not invertible.

5. The rows of that matrix W produce three vectors (I write them as columns):

$$r_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \quad r_2 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \quad r_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1r_1 + y_2r_2 + y_3r_3 = 0$. Find two sets of y 's.

6. Which numbers c give dependent columns so a combination of columns equals zero?

$$\begin{bmatrix} 1 & 1 & 0 \\ 3 & 2 & 1 \\ 7 & 4 & c \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & c \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} c & c & c \\ 2 & 1 & 5 \\ 3 & 3 & 6 \end{bmatrix} \quad \begin{array}{l} \text{maybe} \\ \text{always} \\ \text{independent for } c \neq 0 \end{array}$$

7. If the columns combine into $Ax = 0$, then each of the rows has $r \cdot x = 0$:

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{By rows} \quad \begin{bmatrix} r_1 \cdot x \\ r_2 \cdot x \\ r_3 \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to x ?

8. Moving to a 4×4 difference equation $Ax = b$, find the four components x_1, x_2, x_3, x_4 . Then write this solution as $x = A^{-1}b$ to find the inverse matrix:

$$Ax = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = b.$$

9. What is the cyclic 4 by 4 difference matrix C ? It will have 1 and -1 in each row and each column. Find all solutions $x = (x_1, x_2, x_3, x_4)$ to $Cx = 0$. The four columns of C lie in a “three-dimensional hyperplane” inside four-dimensional space.

10. A forward difference matrix Δ is upper triangular:

$$\Delta z = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3 . What is the inverse matrix in $z = \Delta^{-1}b$?

11. Show that the forward differences $(t+1)^2 - t^2$ are $2t+1 = \text{odd numbers}$. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n , which is _____.

12. The last lines of the Worked Example say that the 4×4 centered difference matrix in (16) is invertible. Solve $Cx = (b_1, b_2, b_3, b_4)$ to find its inverse in $x = C^{-1}b$.

Challenge Problems

13. The very last words say that the 5×5 centered difference matrix is not invertible. Write down the 5 equations $Cx = b$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a “4-dimensional hyperplane” in 5-dimensional space. Hard to visualize.)

14. If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d) . This is surprisingly important: two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent rows, then it also has dependent columns.

2 Chapter 2: Solving Linear Equations

2.1 Vectors and Linear Equations

1. With $A = I$ (the identity matrix) draw the planes in the row picture. Three sides of a box meet at the solution $x = (x, y, z) = (2, 3, 4)$:

$$\begin{array}{lcl} 1x + 0y + 0z = 2 & & \\ 0x + 1y + 0z = 3 & \text{or} & \\ 0x + 0y + 1z = 4 & & \end{array} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Draw the vectors in the column picture. Two times column 1 plus three times column 2 plus four times column 3 equals the right side b .

2. If the equations in Problem 1 are multiplied by 2, 3, 4 they become $Dx = B$:

$$\begin{array}{lcl} 2x + 0y + 0z = 4 & & \\ 0x + 3y + 0z = 9 & \text{or} & \\ 0x + 0y + 4z = 16 & & \end{array} \quad Dx = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix} = B.$$

Why is the row picture the same? Is the solution X the same as x ? What is changed in the column picture—the columns or the right combination to give B ?

3. If equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the vectors in the column picture, the coefficient matrix, the solution? The new equations in Problem 1 would be $x = 2$, $x + y = 5$, $z = 4$.
4. Find a point with $z = 2$ on the intersection line of the planes $x + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$. Find a third point halfway between.
5. The first of these equations plus the second equals the third:

$$\begin{array}{l} x + y + z = 2 \\ x + 2y + z = 3 \\ 2x + 3y + 2z = 5. \end{array}$$

The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also _____. The equations have infinitely many solutions (the whole line L). Find three solutions on L .

6. Move the third plane in Problem 5 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—why not? The first two planes meet along the line L , but the third plane doesn't _____ that line.
7. In Problem 5 the columns are $(1, 1, 2)$ and $(1, 2, 3)$ and $(1, 1, 2)$. This is a “singular case” because the third column is _____. Find two combinations of the columns that give $b = (2, 3, 5)$. This is only possible for $b = (4, 6, c)$ if $c =$ _____.
8. Normally 4 “planes” in 4-dimensional space meet at a _____. Normally 4 column vectors in 4-dimensional space can combine to produce b . What combination of $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 1, 1, 0)$, $(1, 1, 1, 1)$ produces $b = (3, 3, 3, 2)$? What 4 equations for x, y, z, t are you solving?

Problems 9–14 are about multiplying matrices and vectors.

9. Compute each Ax by dot products of the rows with the column vector:

$$\text{(a)} \quad \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \quad \text{(b)} \quad \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

10. Compute each Ax in Problem 9 as a combination of the columns:

$$9(a) \text{ becomes } Ax = 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}.$$

How many separate multiplications for Ax , when the matrix is “3 by 3”?

11. Find the two components of Ax by rows or by columns:

$$\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

12. Multiply A times x to find three components of Ax :

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

13. (a) A matrix with m rows and n columns multiplies a vector with _____ components to produce a vector with _____ components.

(b) The planes from the m equations $Ax = b$ are in _____-dimensional space. The combination of the columns of A is in _____-dimensional space.

14. Write $2x + 3y + z + 5t = 8$ as a matrix A (how many rows?) multiplying the column vector $x = (x, y, z, t)$ to produce b . The solutions x fill a plane or a “hyperplane” in 4-dimensional space. The plane is 3-dimensional with no 4D volume.

Problems 15–22 ask for matrices that act in special ways on vectors.

15. (a) What is the 2×2 identity matrix? I times $\begin{bmatrix} x \\ y \end{bmatrix}$ equals $\begin{bmatrix} x \\ y \end{bmatrix}$.

(b) What is the 2×2 exchange matrix? P times $\begin{bmatrix} x \\ y \end{bmatrix}$ equals $\begin{bmatrix} y \\ x \end{bmatrix}$.

16. (a) What 2×2 matrix R rotates every vector by 90° ? R times $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} -y \\ x \end{bmatrix}$.

(b) What 2×2 matrix R^2 rotates every vector by 180° ?

17. Find the matrix P that multiplies (x, y, z) to give (y, x, z) . Find the matrix Q that multiplies (y, x, z) to bring back (x, y, z) .

18. What 2×2 matrix E subtracts the first component from the second component?
What 3×3 matrix does the same?

$$E \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{and} \quad E \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}.$$

19. What 3×3 matrix E multiplies (x, y, z) to give $(x, y, z + x)$? What matrix E^{-1} multiplies (x, y, z) to give $(x, y, z - x)$? If you multiply $(3, 4, 5)$ by E and then multiply by E^{-1} , the two results are _____ and _____.

20. (a) What 2×2 matrix P_1 projects the vector (x, y) onto the x axis to produce $(x, 0)$?

(b) What matrix P_2 projects onto the y axis to produce $(0, y)$? If you multiply $(5, 7)$ by P_1 and then multiply by P_2 , you get _____ and _____.

21. What 2×2 matrix R rotates every vector through 45° ? The vector $(1, 0)$ goes to $(\sqrt{2}/2, \sqrt{2}/2)$. The vector $(0, 1)$ goes to $(-\sqrt{2}/2, \sqrt{2}/2)$. Those determine the matrix. Draw these particular vectors in the xy plane and find R .

22. Write the dot product of $(1, 4, 5)$ and (x, y, z) as a matrix multiplication Ax . The matrix A has one row. The solutions to $Ax = 0$ lie on a _____ perpendicular to the vector _____. The columns of A are only in _____-dimensional space.

23. In MATLAB notation, write the commands that define this matrix A and the column vectors x and b . What command would test whether or not $Ax = b$?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} 5 \\ -2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$$

24. The MATLAB commands $A = \text{eye}(3)$ and $v = [3; 5]'$ produce the 3×3 identity matrix and the column vector $(3, 4, 5)$. What are the outputs from $A*v$ and $v'*A$? (Computer not needed!) If you ask for $v*A$, what happens?

25. If you multiply the 4×4 all-ones matrix $A = \text{ones}(4)$ and the column vector $v = \text{ones}(4, 1)$, what is Av ? (Computer not needed.) If you multiply $B = \text{eye}(4) + \text{ones}(4)$ times $w = \text{zeros}(4, 1) + 2 * \text{ones}(4, 1)$, what is $B*w$?

26. Draw the row and column pictures for the equations $x - 2y = 0, x + y = 6$.

27. For two linear equations in three unknowns x, y, z , the row picture will show (2 or 3) (lines or planes) in (2 or 3)-dimensional space. The column picture is in (2 or 3)-dimensional space. The solutions normally lie on a _____.

28. For four linear equations in two unknowns x and y , the row picture shows four _____. The column picture is in _____-dimensional space. The equations have no solution unless the vector on the right side is a combination of _____.

29. Start with the vector $u_0 = (1, 0)$. Multiply again and again by the same "Markov matrix"

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}.$$

The next three vectors are u_1, u_2, u_3 :

$$u_1 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .8 \\ .2 \end{bmatrix}, \quad u_2 = Au_1 = \underline{\hspace{2cm}}, \quad u_3 = Au_2 = \underline{\hspace{2cm}}.$$

What property do you notice for all four vectors u_0, u_1, u_2, u_3 ?

Challenge Problems

30. Continue Problem 29 from $u_0 = (1, 0)$ to u_7 , and also from $v_0 = (0, 1)$ to v_7 . What do you notice about u_7 and v_7 ? Here are two MATLAB codes, with `while` and `for`:

<code>u = [1; 0]; A = [.8 .3; .2 .7];</code>	<code>v = [0; 1]; A = [.8 .3; .2 .7];</code>
<code>x = u; k = [0 : 7];</code>	<code>x = v; k = [0 : 7];</code>
<code>while size(x,2) <= 7</code>	<code>for j = 1 : 7</code>
<code> u = A*u; x = [x u];</code>	<code> v = A*v; x = [x v];</code>
<code>end</code>	<code>end</code>
<code>plot(k, x)</code>	<code>plot(k, x)</code>

The u 's and v 's are approaching a steady state vector s . Guess that vector and check that $As = s$. If you start with s , you stay with s .

31. Invent a 3×3 magic matrix M_3 with entries $1, 2, \dots, 9$. All rows and columns and diagonals add to 15. The first row could be $8, 3, 4$. What is M_3 times $(1, 1, 1)$? What is M_4 times $(1, 1, 1, 1)$ if a 4×4 magic matrix has entries $1, \dots, 16$?
32. Suppose u and v are the first two columns of a 3×3 matrix A . Which third columns w would make this matrix singular? Describe a typical column picture of $Ax = b$ in that singular case, and a typical row picture (for a random b).

33. Multiplication by A is a “linear transformation”. Those words mean:

If w is a combination of u and v , then Aw is the same combination of Au and Av .

It is this “linearity” $Aw = cAu + dAv$ that gives us the name “linear algebra”.

Problem: If $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then Au and Av are the columns of A .

Combine $w = cu + dv$. If $w = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$, how is Aw connected to Au and Av ?

34. Start from the four equations $-x_{i+1} + 2x_i - x_{i-1} = i$ (for $i = 1, 2, 3, 4$ with $x_0 = x_5 = 0$). Write those equations in their matrix form $Ax = b$. Can you solve them for x_1, x_2, x_3, x_4 ?

35. A 9×9 Sudoku matrix S has the numbers $1, \dots, 9$ in every row and every column, and in every 3 by 3 block. For the all-ones vector $x = (1, \dots, 1)$, what is Sx ?

A better question is: Which row exchanges will produce another Sudoku matrix? Also, which exchanges of block rows give another Sudoku matrix?

Section 2.7 will look at all possible permutations (reorderings) of the rows. I can see 6 orders for the first 3 rows, all giving Sudoku matrices. Also 6 permutations of the next 3 rows, and of the last 3 rows. And 6 block permutations of the block rows?

2.2 The Idea of Elimination

Problems 1–10 are about elimination on 2×2 systems.

1. What multiple ℓ_{21} of equation 1 should be subtracted from equation 2?

$$\begin{aligned} 2x + 3y &= 1 \\ 10x + 9y &= 11 \end{aligned}$$

After elimination, write down the upper triangular system and circle the two pivots. The numbers 1 and 11 don’t affect the pivots—use them now in back substitution.

2. Solve the triangular system of Problem 1 by back substitution, y before x . Verify that x times $(2, 10)$ plus y times $(3, 9)$ equals $(1, 11)$. If the right side changes to $(4, 44)$, what is the new solution?

3. What multiple of equation 1 should be subtracted from equation 2?

$$\begin{aligned} 2x - 4y &= 6 \\ -x + 5y &= 0 \end{aligned}$$

After this elimination step, solve the triangular system. If the right side changes to $(-6, 0)$, what is the new solution?

4. What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c ?

$$\begin{aligned} ax + by &= f \\ cx + dy &= g \end{aligned}$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y ? The second pivot is missing when $ad = bc$: singular.

5. Choose a right side which gives no solution and another right side which gives infinitely many solutions. What are two of those solutions?

Singular system

$$\begin{aligned} 3x + 2y &= 10 \\ 6x + 4y &= \end{aligned}$$

6. Choose a coefficient b that makes this system singular. Then choose a right side g that makes it solvable. Find two solutions in that singular case.

$$2x + by = 16$$

$$4x + 8y = g$$

7. For which numbers a does elimination break down (1) permanently (2) temporarily?

$$ax + 3y = -3$$

$$4x + 6y = 6$$

Solve for x and y after fixing the temporary breakdown by a row exchange.

8. For which three numbers k does elimination break down? Which is fixed by a row exchange? In each case, is the number of solutions 0 or 1 or ∞ ?

$$kx + 3y = 6$$

$$3x + ky = -6$$

9. What test on b_1 and b_2 decides whether these two equations allow a solution? How many solutions will they have? Draw the column picture for $b = (1, 2)$ and $(1, 0)$.

$$3x - 2y = b_1$$

$$6x - 4y = b_2$$

10. In the xy plane, draw the lines $x + y = 5$ and $x + 2y = 6$ and the equation $y = \underline{\hspace{1cm}}$ that comes from elimination. The line $5x - 4y = c$ will go through the solution of these equations if $c = \underline{\hspace{1cm}}$.

Problems 11–20 study elimination on 3×3 systems (and possible failure).

11. (Recommended) A system of linear equations can't have exactly two solutions. Why?

(a) If (x, y, z) and (X, Y, Z) are two solutions, what is another solution?

(b) If 25 planes meet at two points, where else do they meet?

12. Reduce this system to upper triangular form by two row operations:

$$2x + 3y + z = 8$$

$$4x + 7y + 5z = 20$$

$$-2y + 2z = 0$$

Circle the pivots. Solve by back substitution for z, y, x .

13. Apply elimination (circle the pivots) and back substitution to solve

$$2x - 3y = 3$$

$$4x - 5y + z = 7$$

$$2x - y - 3z = 5$$

List the three row operations: Subtract $\underline{\hspace{1cm}}$ times row $\underline{\hspace{1cm}}$ from row $\underline{\hspace{1cm}}$.

14. Which number d forces a row exchange, and what is the triangular system (not singular) for that d ? Which d makes this system singular (no third pivot)?

$$2x + 5y + z = 0$$

$$4x + dy + z = 2$$

$$y - z = 3$$

15. Which number b leads later to a row exchange? Which b leads to a missing pivot? In that singular case find a nonzero solution x, y, z .

$$x + by = 0$$

$$x - 2y - z = 0$$

$$+ y + z = 0$$

16. (a) Construct a 3×3 system that needs two row exchanges to reach a triangular form and a solution.
 (b) Construct a 3×3 system that needs a row exchange to keep going, but breaks down later.
17. If rows 1 and 2 are the same, how far can you get with elimination (allowing row exchange)? If columns 1 and 2 are the same, which pivot is missing?

$$\begin{array}{ccccc} \text{Equal} & 2x - y + z = 0 & 2x + 2y + z = 0 & & \text{Equal} \\ \text{rows} & 2x - y + z = 0 & 4x + 4y + z = 0 & & \text{columns} \\ & 4x + y + z = 2 & 6x + 6y + z = 2 & & \end{array}$$

18. Construct a 3×3 example that has 9 different coefficients on the left side, but rows 2 and 3 become zero in elimination. How many solutions to your system with $b = (1, 10, 100)$ and how many with $b = (0, 0, 0)$?
19. Which number q makes this system singular and which right side t gives it infinitely many solutions? Find the solution that has $z = 1$.

$$\begin{aligned} x + 4y - 2z &= 1 \\ x + 7y - 6z &= 6 \\ 3y + qz &= t \end{aligned}$$

20. Three planes can fail to have an intersection point, even if no planes are parallel. The system is singular if row 3 of A is a _____ of the first two rows. Find a third equation that can't be solved together with $x + y + z = 0$ and $x - 2y - z = 1$.
21. Find the pivots and the solution for both systems ($Ax = b$ and $Kx = b$):

$$\begin{array}{rclcl} 2x + y & = & 0 & 2x - y & = & 0 \\ x + 2y + z & = & 0 & -x + 2y - z & = & 0 \\ y + 2z + t & = & 0 & -y + 2z - t & = & 0 \\ z + 2t & = & 5 & -z + 2t & = & 5 \end{array}$$

22. If you extend Problem 21 following the 1, 2, 1 pattern or the -1, 2, -1 pattern, what is the fifth pivot? What is the n th pivot? K is my favorite matrix.
23. If elimination leads to $x + y = 1$ and $2y = 3$, find three possible original problems.
24. For which two numbers a will elimination fail on $A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$?
25. For which three numbers a will elimination fail to give three pivots?

$$A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix} \text{ is singular for three values of } a.$$

26. Look for a matrix that has row sums 4 and 8, and column sums 2 and s :

$$\text{Matrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \begin{cases} a + b = 4 \\ a + c = 2 \\ c + d = 8 \\ b + d = s \end{cases}$$

The four equations are solvable only if $s = \underline{\hspace{1cm}}$. Then find two different matrices that have the correct row and column sums. *Extra credit:* Write down the 4×4 system $Ax = b$ with $x = (a, b, c, d)$ and make A triangular by elimination.

27. Elimination in the usual order gives what matrix U and what solution to this “lower triangular” system? We are really solving by *forward substitution*:

$$\begin{aligned} 3x &= 3 \\ 6x + 2y &= 8 \\ 9x - 2y + z &= 9 \end{aligned}$$

28. Create a MATLAB command $A(2, :) = \dots$ for the new row 2, to subtract 3 times row 1 from the existing row 2 if the matrix A is already known.

Challenge Problems

29. Find experimentally the average 1st and 2nd and 3rd pivot sizes from MATLAB’s $[L, U] = \text{lu}(\text{rand}(3))$. The average size $\text{abs}(U(1, 1))$ is above $\frac{1}{2}$ because lu picks the largest available pivot in column 1. Here $A = \text{rand}(3)$ has random entries between 0 and 1.
30. If the last corner entry is $A(5, 5) = 11$ and the last pivot of A is $U(5, 5) = 4$, what different entry $A(5, 5)$ would have made A singular?
31. Suppose elimination takes A to U without row exchanges. Then row j of U is a combination of which rows of A ? If $Ax = 0$, is $Ux = 0$? If $Ax = b$, is $Ux = b$? If A starts out lower triangular, what is the upper triangular U ?
32. Start with 100 equations $Ax = 0$ for 100 unknowns $x = (x_1, \dots, x_{100})$. Suppose elimination reduces the 100th equation to $0 = 0$, so the system is “singular”.
- Elimination takes linear combinations of the rows. So this singular system has the singular property: Some linear combination of the 100 rows is _____.
 - Singular systems $Ax = 0$ have infinitely many solutions. This means that some linear combination of the 100 _____ is _____.
 - Invent a 100 by 100 singular matrix with no zero entries.
 - For your matrix, describe in words the row picture and the column picture of $Ax = 0$. Not necessary to draw 100-dimensional space.

2.3 Elimination Using Matrices

Challenge Problems

2.4 Rules for Matrix Operations

Challenge Problems

2.5 Inverse Matrices

Challenge Problems

2.6 Elimination = Factorization: $A = LU$

Challenge Problems

2.7 Transposes and Permutations

Challenge Problems

3 Chapter 3: Vector Spaces and Subspaces

3.1 Spaces of Vectors

Challenge Problems

3.2 The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$

Challenge Problems

3.3 The Complete Solution to $Ax = b$

Challenge Problems

3.4 Independence, Basis and Dimension

Challenge Problems

3.5 Dimensions of the Four Subspaces

Challenge Problems

4 Chapter 4: Orthogonality

4.1 Orthogonality of the Four Subspaces

Challenge Problems

4.2 Projections

Challenge Problems

4.3 Least Squares Approximations

Challenge Problems

4.4 Orthonormal Bases and Gram-Schmidt

Challenge Problems

5 Chapter 5: Determinants

5.1 The Properties of Determinants

Challenge Problems

5.2 Permutations and Cofactors

Challenge Problems

5.3 Cramer's Rule, Inverses, and Volumes

Challenge Problems

6 Chapter 6: Eigenvalues and Eigenvectors

6.1 Introduction to Eigenvalues

Challenge Problems

6.2 Diagonalizing a Matrix

Challenge Problems

6.3 Systems of Differential Equations

Challenge Problems

6.4 Symmetric Matrices

Challenge Problems

6.5 Positive Definite Matrices

Challenge Problems

7 Chapter 7: The Singular Value Decomposition (SVD)

7.1 Image Processing by Linear Algebra

Challenge Problems

7.2 Bases and Matrices in the SVD

Challenge Problems

7.3 Principal Component Analysis (PCA by the SVD)

Challenge Problems

7.4 The Geometry of the SVD

Challenge Problems